

6	(a) lines perpendicular to surface or lines are radial	M1	
	lines <u>appear</u> to come from centre	A1	[2]
(b)	(i) $F_E = (1.6 \times 10^{-19})^2 / 4\pi\epsilon_0 x^2$	C1	
	$F_G = G \times (1.67 \times 10^{-27})^2 / x^2$	C1	
	$F_E / F_G = (1.6 \times 10^{-19})^2 \times (8.99 \times 10^9) / [(1.67 \times 10^{-27})^2 \times (6.67 \times 10^{-11})]$ $= 1.2 (1.24) \times 10^{36}$	A1	[3]
	(ii) $F_E \gg F_G$	B1	[1]

Page 5	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2016	9702	42